

Alan (Shixuan) Zhao

sxuan@stanford.edu | sxuan.alan@gmail.com | [linkedin.com/in/shixuan-z-3074aa384](https://www.linkedin.com/in/shixuan-z-3074aa384) | [YOUR-WEBSITE](#)

EDUCATION

Stanford University <i>M.S. in Mechanical Engineering (Distinction in Research)</i>	Sep. 2025 – Present GPA: 3.86/4.0
The University of Arizona <i>B.S. in Mechanical Engineering, Summa Cum Laude</i>	Sep. 2021 – July 2025 GPA: 4.0/4.0
Hebei University of Technology <i>B.Eng. in Mechanical Engineering (Dual Degree), Outstanding Graduate</i>	Sep. 2021 – July 2025 GPA: 3.95/4.0 (Rank 2/73)

EXPERIENCE

Graduate Research Assistant <i>Biomimetics & Dexterous Manipulation Lab (BDML), Stanford University</i> <i>Research Intern, Stanford Robotics Center</i> Project: Humanoid Wall Climbing with Compliant Spines	Jan. 2026 – Present PI: Prof. Mark Cutkosky June 2026 – Sep. 2026 Co-advised by Prof. Karen Liu
Contact Modeling: Built a physics-grounded micro-spine contact model in MjLab (MuJoCo): a compliant spring-unit array with a limit-surface failure criterion on a passive gimbal end-effector. Reference-Guided RL: Synthesized whole-body climbing references with an LLM agent via contact-constrained optimization and force-feasibility certificates; trained PPO tracking policies over a 45–90° wall-inclination curriculum. Proprioceptive Adaptation: Distilled privileged contact states into an RMA-style estimator over proprioceptive history, removing the need for load sensors on the end-effector. Sim-to-Real Deployment: Deployed policies on the physical humanoid via onboard C++, identified micro-spine end-effector failure modes, and am redesigning the gripper and gimbal guided by simulated wrench analysis.	
Project: Fluorescent Whisker Array for Underwater Source Localization	Submitted to ICRA 2027
CNN Decoding: Built a lightweight CNN that decodes 2s of whisker deformation into source position, achieving 88 mm spatial and 2.5° bearing error and real-time tracking of moving sources.	
Project: Lego-Like Stiffness Configuration for Compliant Robotic Wrists	Submitted to ICRA 2027
Robot Control: Implemented robot-arm control for wrist-obstacle contact experiments, including 1 m/s trials.	
Research Intern <i>Institute of Automation, Chinese Academy of Sciences (CASIA)</i>	July 2024 – Sep. 2024 PI: Dr. Xiaohu Zhou
- Designed a motor-imagery paradigm and EEG processing pipeline for a non-invasive brain-computer interface. - Developed multi-process online BCI programs to control ping-pong games in real time.	
Undergraduate Research Assistant <i>Hebei University of Technology</i>	Sep. 2022 – Aug. 2023 Supervisors: Prof. Lixin Sun, Prof. E Cheng
Designed RoboBLUE, a campus welcome robot: chassis design, 3D prototyping, and Omron PLC programming.	

PROJECTS

Senior Design: Efficient EEG Decoding for Real-Time BCI <i>Python, PyTorch</i>	Jan. 2025 – June 2025
Designed a lightweight CNN combining a Dilated-Inception Block for multi-scale features, Efficient Channel-Temporal Attention, and Ghost Modules to reduce model cost.	

TECHNICAL SKILLS

Robot Learning & Control: Deep Reinforcement Learning, Imitation Learning, Contact Modeling
Simulation & Frameworks: MuJoCo / MjLab, Isaac Gym, Isaac Sim, PyTorch, ROS 2, OpenCV
Programming: Python, C/C++, MATLAB
Hardware & Design: SolidWorks, ANSYS, ABAQUS, 3D Printing, STM32, Arduino, Raspberry Pi